

A Python Programmed Resnet-50 CNN Designed for Low Light Image Enhancement

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Abstract—Enhancing the quality of photos taken in low light is the goal of the computer vision problem known as Low-Light Image Enhancement. Increasing the brightness, clarity, and visual attractiveness of images while minimizing noise and distortion is the aim of low-light image enhancement. Most CNN-based methods have complex network structures and the network consists of several sub-networks. The challenges posed by insufficient illumination in image enhancement with their visual quality low light condition often lead to reduced visibility and loss of important details, impacting various applications in autonomous system. In contrast to these problems we utilize ResNet-50 for image segmentation up to 50 layers to reduce noises in image, scattering of images and more focusing on higher brightness places in an image. Python programming is used to restoring edges and textures and improving overall image visibility. According to our experimental results, the proposed method achieves improvised image in clarity and environmental issues such as fog, pollution.

Keywords—Image enhancement, CNN-Efficient Net B0, SSIM loss, Artificial Intelligence, gradient vanishing, Enhancement methods

I. INTRODUCTION

Although they can learn intricate mappings between high-quality and low-light pictures, CNNs a well-liked option for this problem is ResNet-50, a deep CNN architecture with 50 layers. Using the learned features of a pre-trained ResNet-50 model that has been refined on a dataset of low-light photos, the visual quality of low-light photos will be enhanced.

Our goal in this study is to create a Python application that enhances low-light images using a pre-trained ResNet-50 CNN. The framework will be trained on a dataset of low-light photos to learn how to improve their visual quality. The application will be implemented in Python and will assess performance using image quality metrics such as Structural Similarity Index (SSIM) and Peak Signal-to-Noise Ratio (PSNR).

Low-light image enhancement is used in a variety of situations, such as fog dehazing, underwater imaging, and night-day time conversions. All of these circumstances work with photos that have backlighting, uneven lighting, and low lighting. Their goal is to create perceptual images from these poorly obtained ones. All of the Low Light Image Enhancement works fall under the areas listed below. Resnet-

50 uses a variety of techniques, including Histogram Equalization, Retinex Decomposition, and Deep Learning, to transform low-light images into brighter, more enhanced images.

Deep learning methods have recently been applied extensively and shown to have great accuracy in predicting images in natural light. One kind of artificial intelligence that draws inspiration from the human brain is deep learning. It analyzes vast volumes of data using sophisticated neural networks, revealing the complex patterns and relationships depicted in figure 1. This pushes the limits of what robots can accomplish by enabling them to carry out activities like speech translation, image identification, and even the creation of creative material.

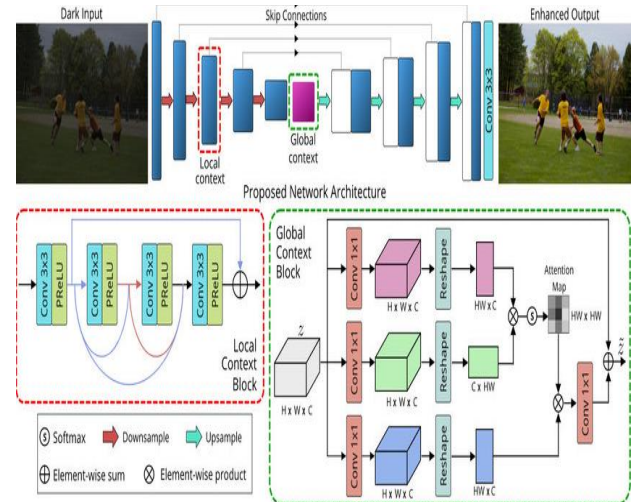


Fig.1. Block Diagram network architecture

They are categorized according to their learning input as

- Supervised learning
- Unsupervised learning

In supervised learning, a machine learning model is trained with marked data. This information includes input features and the intended outputs that go along with them. In essence, the model picks up a mapping function between these outputs and characteristics. For problems including object

detection, regression, and classification, supervised learning is frequently employed.

The machine learning model in unsupervised learning works with unlabeled data, which is data that does not have any predetermined labels or groups. In contrast to input-output pairings in supervised learning, unsupervised learning algorithms seek to uncover hidden patterns and structures in the data. For tasks including anomaly detection, dimensionality reduction, and clustering, unsupervised learning is frequently employed.

II. EXISTING SYSTEM

Inadequate illumination can have an impact on photos that are taken in low light. A digital image processing method called low-light image enhancement, or LLIE, transforms low-light photographs into typical, natural-light ones. LLIE's three primary goals are to replicate natural brightness and colour, preserve edges and textures, and minimize noise and artifacts. Deep learning-based methods that can achieve high performance have been the focus of numerous sorts of research in recent years. The lengthy inference time caused by intricate network topologies is the main issue with these approaches, though.

In order to address the trade-off between implementation time and performance, they present a straightforward network with efficient modules. To maintain edges and textures, pre-processing is used and we use a U-Net structure. To preserve high-frequency details, our method integrates Pixel Shuffler, while Channel Attention mechanisms help enhance color fidelity and illumination. Additionally, we incorporate a Res FFT-ReLU block to effectively suppress noise. Experimental evaluations demonstrate that our approach surpasses traditional low-light image enhancement (LLIE) methods in both speed and overall performance.

By controlling the effect of classic histogram equalization (HE), the dynamic histogram equalization (DHE) technology enhances an image without sacrificing any of its nuances. Dynamic Histogram Equalization (DHE) improves contrast by segmenting the image histogram at local minima and then performing histogram equalization independently within each segment, allowing better preservation of image details. To make sure there are no dominating sections, these partitions are also subjected to a repartitioning test. This technique performs better than other current methods because it improves contrast without producing unwanted artifacts or serious negative effects like washed-out appearance or checkerboard effects.

This study investigates the optimal skid height and landing safety of an aerial vehicle operating on sloped terrain by analyzing the surface angle of the landscape. Advanced computer vision techniques such as Canny edge detection and Haar cascades are employed to identify objects and extract landscape boundaries from captured images. The redesigned landing skid aims to enhance the aerial vehicle's ability to land securely and efficiently on uneven ground.

Typically, little information is available to improve photos taken in poor light. To obtain trustworthy information for the augmentation, we suggest taking advantage of the interactions between superposed luminance levels. A noise-resilient transformation is then used after Grover's method executes the interaction that causes information to flow between brightness levels. Based on both qualitative and quantitative data, our

concept improves low light images better than the state-of-the-art.

III. PROPOSED METHODOLOGY

Although photos taken in low light have poor visual quality, enhancing photographs in low light is a major difficulty. We address this by treating low-light enhancement as a residual learning problem, with the goal of estimating the difference (residual) between photos taken in normal light and those taken in low light. In order to improve low-light picture enhancement, we present a unique Convolutional Neural Network (CNN) based on the ResNet50 architecture in this study.

Although it is classified as a low-level image processing activity, low-light enhancement is very different from tasks like image denoising and super resolution. Pixel values in deteriorated photos are usually near to their true values in super resolution and denoising, with just slight variations in average pixel values. Low-light enhancement, on the other hand, results in images with higher noise and less contrast. As a result, we create a customized ResNet50 architecture for improving low-light photos, emphasizing the acquisition of complex low-light-specific properties.

A. Algorithm

CNNs, such as the ResNet-50 architecture, are widely used for object classification tasks involving multiple categories. Rather than relying on clustering, these models learn complex hierarchical features through convolutional layers. CNNs are often considered translation-invariant due to their shared-weight structure and receptive field mechanism. Their versatility makes them suitable for a range of tasks including image and video recognition, medical imaging, natural language processing, and recommendation systems.

Convolutional Neural Networks (CNNs) can be viewed as a regularized form of multilayer perceptrons (MLPs). While MLPs are typically fully connected—where each neuron in one layer connects to every neuron in the next—this full connectivity often leads to over fitting. Traditional regularization techniques such as weight decay and dropout are commonly applied to mitigate this. CNNs, however, achieve a different form of regularization by leveraging local connectivity and shared weights. They capture hierarchical patterns in data by learning simple features in early layers and progressively combining them into more complex representations. As a result, CNNs exhibit lower parameter complexity and reduced risk of over fitting compared to fully connected networks.

Convolutional neural networks (CNNs) are inspired by biological processes, particularly the visual perception mechanisms found in animals. Although the precise correspondence between artificial neuron connections and the animal visual cortex is not fully understood, CNNs mimic the concept of receptive fields. In the visual cortex, a receptive field refers to the specific region of the input space where a neuron responds. These fields often overlap, enabling the full visual field to be represented through the combined activity of many neurons.

CNNs require a relatively lower amount of pre-processing than other image classification techniques. In conventional systems, the convolution kernels or filters are manually built; this suggests that the network learns to optimize them. One

significant benefit is its independence from human interaction in feature extraction and prior knowledge.

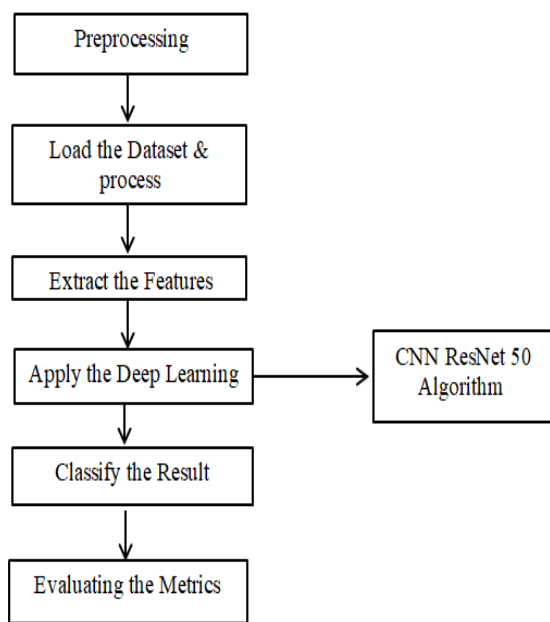


Fig. 2. Flow chart

A shape-based tensor from CNN based input,

$$\begin{bmatrix} \text{Number of} \\ \text{images} \end{bmatrix} \times \begin{bmatrix} \text{image} \\ \text{height} \end{bmatrix} \times \begin{bmatrix} \text{input} \\ \text{Channels} \end{bmatrix} \quad (1)$$

The picture is abstracted to a feature map with shape after going through a convolutional layer

$$\begin{bmatrix} \text{NUMBER OF} \\ \text{IMAGES} \end{bmatrix} \times \begin{bmatrix} \text{FEATURE} \\ \text{IMAGE} \\ \text{HEIGHT} \end{bmatrix} \times \begin{bmatrix} \text{FEATURE} \\ \text{MAP} \\ \text{WIDTH} \end{bmatrix} \times \begin{bmatrix} \text{FEATURE} \\ \text{MAP} \\ \text{CHANNELS} \end{bmatrix} \quad (2)$$

A neural network's convolutional layer ought to possess the following qualities:

Convolutional filters in CNNs are defined by several hyperparameters, including their height, width, and the number of input and output channels. One key constraint is that the depth of the filter (i.e., the number of input channels) must match the depth of the input feature map. This ensures that each filter processes all available channels of the input data correctly.

The convolution operation in neural networks relies on key hyperparameters such as padding and stride. After applying the convolution, each layer passes the resulting feature maps to the next. This mechanism mimics how neurons in the visual cortex respond to specific regions of the visual field—referred to as receptive fields. Each convolutional neuron processes only the input within its receptive region, enabling localized feature extraction. Unlike fully connected networks, which are parameter-heavy and inefficient for image data, convolutional networks are more suitable for processing photographs due to their spatial locality and weight sharing.

Processing image data with fully connected networks is inefficient due to the high dimensionality—each pixel in an image represents a unique input feature. For instance, a fully

connected layer handling a 100×100 pixel image would require 10,000 weights per neuron, resulting in a massive number of parameters. Convolutional layers address this by significantly reducing parameter count through local connectivity and weight sharing. A 5×5 filter, for example, requires just 25 learnable weights regardless of the image size, which makes training more efficient. Additionally, using fewer parameters helps reduce overfitting and mitigates issues like vanishing or exploding gradients during backpropagation.

To reduce computational complexity, convolutional neural networks often use pooling layers—either local or global. These layers downsample feature maps by aggregating information from small regions, effectively reducing spatial dimensions. Local pooling, such as 2×2 max or average pooling, combines nearby activations into a single representative value, which helps preserve key features while lowering the data size passed to the next layer.

Global pooling operates across all spatial positions in a feature map, summarizing the output of an entire convolutional layer. The two most commonly used pooling methods are average pooling, which computes the mean of values within a region, and max pooling, which selects the highest value. In convolutional neural networks, the convolutional layer plays a fundamental role. It consists of trainable filters (or kernels) that slide across the input volume with a limited spatial extent, while spanning the full depth of the input data.

During the forward pass in a convolutional neural network, each filter slides over the spatial dimensions of the input volume. At each position, the filter computes a dot product with the overlapping input region, generating a corresponding value in the resulting 2D feature map, also known as the activation map.

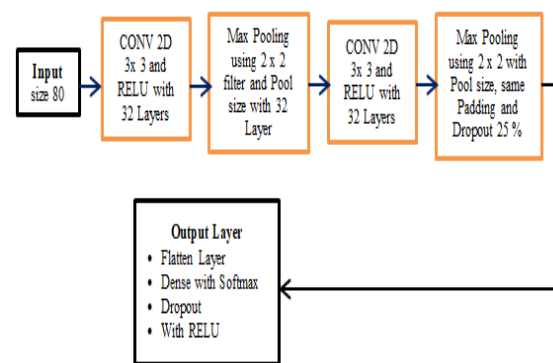


Fig. 3. Feature Extraction Process

In a convolutional neural network, each filter is trained to respond to specific features at particular spatial locations within the input. The output of a convolutional layer consists of multiple feature maps, one per filter, which are stacked along the depth axis to form the final output volume. Each neuron in a feature map processes a local region of the input and shares its weights with others in the same map, allowing efficient feature extraction across the spatial dimensions.

B. Proposed Methodology CNN with RESNET 50

Residual Networks (ResNets) are a type of convolutional neural network (CNN) designed to address the vanishing gradient problem in deep architectures.

Widely used in computer vision tasks, ResNets introduce shortcut connections that allow gradients to flow more effectively during training. ResNet-50, a popular variant, consists of 50 layers including 48 convolutional layers, one max pooling layer, and one average pooling layer. These layers are organized into residual blocks that enable the construction of very deep networks without degradation in performance.

ResNet-34 is part of the family of residual neural networks and includes 34 learnable layers. These models are often pretrained on large-scale datasets such as ImageNet, which contains over a million labeled images across various categories. A key innovation in ResNet is the use of shortcut (or skip) connections, which help mitigate the vanishing gradient problem by allowing the network to learn residual mappings. Instead of learning the full transformation, each block learns the difference—or residual—between the input and the desired output, enabling the training of much deeper networks.

VGG architectures such as VGG-16 and VGG-19 utilize uniform 3×3 convolutional filters and were influential in early deep learning models for image recognition. However, ResNets are more computationally efficient. For example, while VGG-19 requires approximately 19.6 billion floating-point operations (FLOPs), a ResNet-34 achieves similar performance with only 3.6 billion FLOPs, and ResNet-18 requires just 1.8 billion FLOPs. This significant reduction in computation makes ResNets faster and more scalable for deeper architectures.

ResNet-50 comprises fifty levels and five blocks, each of which has a set of residual blocks. The residual blocks allow information from earlier layers to be preserved, which helps the network create better representations of the data it receives. ResNet is what you ought to use. It is deep enough, with 34, 50, or 101 layers, to begin. As the hierarchy's depth increases, so do the classification accuracy and representation capabilities.

The comparatively shallow CNN is unable to outperform random guessing (which has a 50% chance of being correct when guessing between two classes) when trained on just 20 photos per class. ResNet50 is a deep pre-trained network that requires only 20 images per class and has an accuracy rate of 70%.

IV. PROPOSED WORK

Though low-light enhancement is categorized as a low-level image processing activity, it is not the same as activities like image denoising and super resolution. Pixel values in degraded photos are usually near to their true values in super resolution and denoising, with just slight variations in average pixel values. In contrast, low-light enhancement causes images to have more noise and less contrast. Therefore, we create a customized ResNet50 architecture for improving low-light photos, emphasizing the acquisition of complex low-light-specific properties.

We aim to enhance the visual quality of low-light images by training a ResNet50-based convolutional neural network

on a specialized dataset. The model is designed to learn and adapt to the unique characteristics of low-light environments. This approach has practical relevance across various fields, including digital photography, video surveillance, and autonomous vehicle systems, where improving visibility under challenging lighting conditions is critical.

By training our CNN based on ResNet50 on a dataset of low-light photos, we hope to improve the visual quality of the images by efficiently comprehending and utilizing the special qualities of low-light conditions. With potential uses in a number of domains, including photography, surveillance, and autonomous driving, this method provides a practical way to enhance image quality in low light.

A. General formulation of low-light image enhancement

Over the past few decades, the science of computer vision has developed a variety of methods for enhancing low-light images. The notion that intrinsic properties of underlying contents (such as object reflectance, background colour, noise level, etc.) should be maintained during the enhancement process is shared by the majority of earlier methods, despite the fact that their approaches to algorithm design differ greatly from one another. The general steps involved in improving low-light images can be expressed simply as follows:

$$E(x, y) = f(f(x, y)) \quad (3)$$

B. Implementation

The Drive was interfaced with Colab, and all required libraries—including PyTorch, TensorFlow, Wandb, Matplotlib, among others—were imported.

To access the images, the Custom Dataset class needs to be constructed; to increase the quantity of training data, the Data Augmentation class needs to be defined. In addition to this, establish the learning parameters and start training. Following training, pictures from the Evaluation dataset are used to confirm the Model's performance.

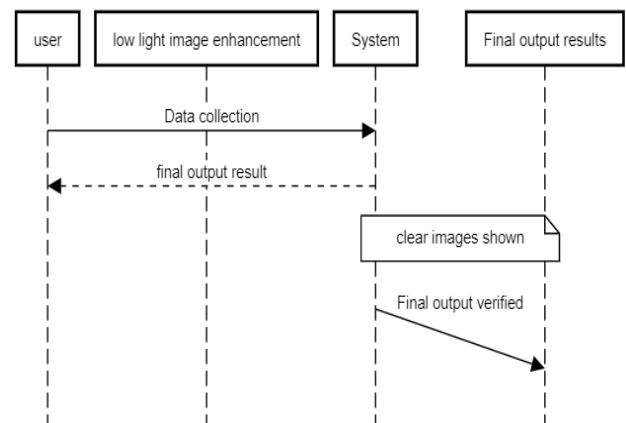


Fig.4 .Low light image enhancement system

V. RESULTS VALIDATION

The proposed CNN with ResNet-50 approach for low-light picture enhancement shows impressive effectiveness in improving the quality of photos taken in difficult lighting situations. Our approach uses a specific module to leverage multi-scale characteristic maps and SSIM degradation to preserve image textures, and it beats previous methods in adaptively enhancing the contrast of low-light photos.

Experiments reveal notable enhancements in brightness, clarity, and visual appeal without adding undue noise or distortion, confirming the superiority of our approach. Our method produces an adaptively improved contrast that not only improves image quality but also performs better than other contrast enhancement methods.

Overall, our CNN with ResNet-50 based approach's efficacy shows that it can be a powerful solution for problems involving the augmentation of low-light images, offering practical benefits in a variety of computer vision applications.



Fig.5. Low light input image

A. Results

The low light input image in Figure 5 is processed using machine learning to save processing time and convert the image to PNG format. Google Colab is used to launch the image, which is then saved on a drive.

B. Parameters

Model: "functional_1"

Layer (type)	Output Shape	Param #
input_1 (InputLayer)	[(None, None, None, 1)]	0
conv2d (Conv2D)	(None, None, None, 64)	1664
conv2d_1 (Conv2D)	(None, None, None, 64)	36928
conv2d_2 (Conv2D)	(None, None, None, 32)	18464
conv2d_3 (Conv2D)	(None, None, None, 9)	2601
tf_op_layer_DepthToSpace (TensorFlowOpLayer)	(None, None, None, 1)	0

=====
Total params: 59657 (233.04 KB)
Trainable params: 59657 (233.04 KB)
Non-trainable params: 0 (0.00 Byte)

Fig.6. Functional parameter

1/1 [=====] - 0s 200ms/step
PSNR of low resolution image and high resolution image is 20.9315
PSNR of predict and high resolution is 21.3451

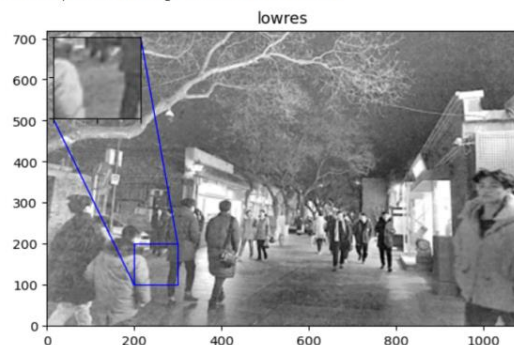


Fig.7. Low resolution image

Low resolution is the term used to describe a display, video, or image that has less clarity or detail. It indicates that,

in comparison to its high-resolution contenders, the content appears less crisp and precise. This may be because there are fewer pixels or a smaller colour scheme used to depict the image. The low resolution image of the trained input image is displayed in Fig. 7.

When compared to low-resolution photos, high-resolution images have more information and clarity since they have a lot more pixels per unit of area. Pixels per inch (PPI) or dots per inch (DPI) are commonly used to measure resolution.

By changing the image resolution from high to low and low to high the contents of the images at its sharp edges are enhanced to give a better image.

Predicted image is images which automatically fill the empty places in the image and increase or decrease the contrast, brightness value of image in required pixels. Fig. 9 shows the predicted resolution image of the trained input image after the higher resolution training.

The LLIE technique based on CNN and ResNet-50 shows impressive effectiveness in improving the quality of photos taken in difficult lighting situations. Results from experiments confirm the effectiveness of our approach, demonstrating notable enhancements in brightness, clarity, and aesthetic appeal without adding undue distortion or noise.

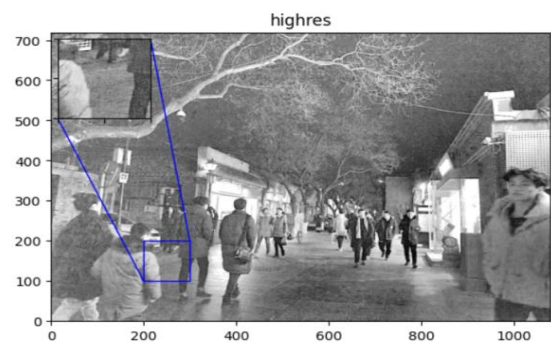


Fig.8 .High resolution image

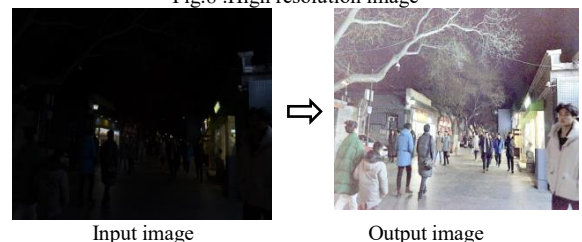


Fig.9 .Final predicted image

VI. CONCLUSION

In this work, we proposes a CNN with ResNet-50 architecture for enhancing low-light images, utilizing multiscale feature maps and SSIM loss for adaptive contrast enhancement while preserving textures.

Implemented in Python and Google Colab Notebook, the method demonstrates superior performance over traditional techniques, leveraging ResNet-50's depth and hierarchical representation for real-world applications, advancing computer vision, and fostering future AI and ML innovations.

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